

Athena assignment

For this assignment I created a separate bucket



The screenshot shows the AWS Glue console interface. At the top, there is a text input field containing 'catalog.'. Below it, the 'Bucket name' section is highlighted with a blue border. The 'Bucket name' field contains the text 'ana-athenalab'. To the right of the field are three buttons: a dropdown menu with 'on' and a downward arrow, a 'Run' button, and a prominent orange 'Create crawler' button. Below the 'Bucket name' field, a small grey text note states: 'Bucket names must be 3 to 63 characters and unique wit'.

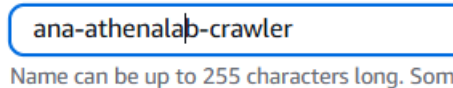
I proceeded to create a new crawler named:

`ana-athenalab-crawler`

A **Glue Crawler** is used to scan data stored in places such as Amazon S3 and discover the structure of the data. In this case, the crawler needed to discover the structure of the CSV file I uploaded. The crawler does **not move the data**. It only examines it and creates metadata about it.

Crawler details [Info](#)

Name



The screenshot shows the 'Name' field in the AWS Glue console. The field contains the text 'ana-athenalab-crawler'. Below the field, a small grey text note states: 'Name can be up to 255 characters long. Som'.

I selected **Not yet** at the data source configuration. This was the correct option because my S3 data had not yet been linked into a Glue Data Catalog table.

I added an **S3 data source**.

`s3://ana-athenalab/athenalab/revenue...`

This told Glue where the CSV file was stored.

So at this point:

S3 = stores the actual CSV data

Glue Crawler = reads the structure of that data

Data source configuration

Do you have data already mapped to Glue tables?

☒ Not yet

Select one or more data sources to be crawled.

☐ Yes

Select existing tables from your Glue Data Catalog.

Data sources (0) [Info](#)

This is the list of data sources to be scanned by the crawler.

Edit

Remove

Add a data source

Type	Data source	Parameters
------	-------------	------------

You don't have any data sources.

Add a data source

I initially created/selected:

AWSGlueServiceRole-Athenalab-ana

The IAM role is important because Glue needs permission to:

- access the S3 bucket
- read the CSV file
- access the Glue Data Catalog
- create/update metadata

Without the IAM role, Glue cannot access the resources it needs.

☐ In a different account

S3 path

Browse for or enter an existing S3 path.

[View](#)

[Browse S3](#)

All folders and files contained in the S3 path are crawled. For example, type s3://MyBucket/MyFolder/ to crawl all objects in MyFolder within MyBucket.

Role name

Maximum 64 characters. Use alphanumeric and +,=,_,.

Output configuration [Info](#)

Target database

[Clear selection](#)

[Add database](#)

My first crawler creation failed because AWS Glue was unable to assume the IAM role I selected. The role's trust policy was not configured correctly for the Glue service. I rectified this by selecting an existing Glue service role that had the correct permissions and trust relationship.

Step 1: Set crawler properties

Set crawler properties

Name	Description
ana-athenalab-crawler	-

Step 2: Choose data sources and classifiers

Data sources (1) [Info](#)

The list of data sources to be scanned by the crawler.

Type	Data source
S3	s3://ana-athenalab/athena

Step 3: Configure security settings

Configure security settings

IAM role	Security configuration
AWSGlueServiceRole-Athenalab-ana	-

Step 4: Set output and scheduling

Set output and scheduling

Database	Table prefix - <i>optional</i>
ana_db_revenue	-

I selected:

ana_db_revenue

A **Glue Database** is not a database containing the actual CSV data.

Before creating it, AWS showed the configuration:

- Crawler: ana-athenalab-crawler
- Data source: S3
- IAM role

- Database: ana_db_revenue
- No table prefix

This review page allowed me to verify the crawler settings before creating it.

After correcting the IAM role, the crawler was created successfully.

❌ **One crawler failed to create**
 The following crawler failed to create: "ana-athenalab-crawler"
 Here is the most recent error message: Service is unable to assume provided role. Please verify role's TrustPolicy

Crawler properties

Name ana-athenalab-crawler	IAM role AWSGlueServiceRole-Ana-salesproject	Database ana_db_revenue	State READY
Description -	Security configuration -	Lake Formation configuration -	Table prefix -
Maximum table threshold -			

▶ Advanced settings

I then selected **Run crawler**.

The crawler started successfully and eventually showed:

Status: Completed The screenshot is showing a duration of approximately **53 seconds**.

So Glue scanned the S3 data, while the crawler is examining things such as file format, delimiter, header, column names, column data types, S3 location.

✅ **Crawler successfully starting**
 The following crawler is now starting: "ana-athenalab-crawler"

	Start time (UTC) ▲	End time (UTC) ▼	Current/last duration ▼	Status ▼	DPU hours
☐	August 12, 2026 at 09:48:39	August 12, 2026 at 09:49:33	53 s	✅ Completed	

After the crawler completed, I have checked the Glue Data Catalog. In which I found inside ana_db_revenue, revenue_csv, which proved that the crawler worked, so glue classified the file as CSV and linked the table to the S3 location containing the data.

Tables (1)

Last updated: August 12, 2026

View and manage all available tables.

Choose catalog

743737183475

Choose database

ana_db_revenue

Filter tables

☐	Name	Database	Location	Classification	Deprecated
☐	revenue_csv	ana_db_revenue	s3://ana-athenalab/atf	CSV	-

After opening the schema, I've seen three columns: year, revenue, profits. This is one of the benefits of crawler, we do not have to manually define every column.

Schema

Partitions

Indexes

Column statistics - new

Schema (3)

View and manage the table schema.

Filter schemas

#	Column name	Data type
1	year	bigint
2	revenue	bigint
3	profits	bigint

Database

ana_db_revenue

Tables and views

Create

Filter tables and views

▼ Tables (1)

< 1 >

+ revenue_csv

⋮

► Views (0)

< 1 >

Data source configuration

Is your data already mapped to Glue tables?

☒ Not yet
Select one or more data sources to be crawled.

☐ Yes
Select existing Glue tables

Data sources (1) Info

The list of data sources to be scanned by the crawler.

Type	Data source
☐ S3	s3://ana-athenalab/athenalab/

Choose database
ana_db_revenue

Filter tables

☐	Name	Database	Location	Classification	Deprecated	View data
☐	revenue_csv	ana_db_revenue	s3://ana-athenalab/athena	CSV	-	Table data

Query 2

1	SHOW CREATE TABLE revenue_csv;	
---	--------------------------------	--

SQL Ln 1, Col 31

Run again

▼

Explain

Cancel

Clear

Create ▼

Query results

Query status

Completed

```
CREATE EXTERNAL TABLE `revenue_csv` (  
  `year` bigint,  
  `revenue` bigint,  
  `profits` bigint)  
ROW FORMAT DELIMITED  
  FIELDS TERMINATED BY ','  
STORED AS INPUTFORMAT  
  'org.apache.hadoop.mapred.TextInputFormat'  
OUTPUTFORMAT
```

✓ Completed

```
CREATE EXTERNAL TABLE `revenue_csv` (  
  `year` bigint,  
  `revenue` bigint,  
  `profits` bigint)  
ROW FORMAT DELIMITED  
  FIELDS TERMINATED BY ','  
STORED AS INPUTFORMAT  
  'org.apache.hadoop.mapred.TextInputFormat'  
OUTPUTFORMAT  
  'org.apache.hadoop.hive.ql.io.HiveIgnoreKeyTextOutputFormat'  
LOCATION  
  's3://ana-athenalab/athenalab/revenue.csv'  
TBLPROPERTIES (  
  'CRAWL_RUN_ID'='58c8eb9f-f175-4b43-aad5-a84b8d84d08c',  
  'CrawlerSchemaDeserializerVersion'='1.0',  
  'CrawlerSchemaSerializerVersion'='1.0',  
  'UPDATED_BY_CRAWLER'='ana-athenalab-crawler',  
  'areColumnsQuoted'='false',  
  'averageRecordSize'='18',  
  'classification'='csv',  
  'columnsOrdered'='true',  
  'compressionType'='none',  
  'delimiter'=',',  
  'objectCount'='1',  
  'recordCount'='11',  
  'sizeKey'='198',  
  'skip.header.line.count'='1',  
  'typeOfData'='file')
```

✓ Query 2 : ✕ | ✓ Query 3 : ✕

```
1 ALTER TABLE revenue_csv  
2 SET LOCATION 's3://ana-athenalab/athenalab/';
```

SQL Ln 1, Col 1

Run again

Explain [↗](#)

Cancel

Clear

Create ▾

Query results

Query status

✓ Completed

Query successful.

✓ Query 2 : ✕ | ✓ Query 3 : ✕

```
1 SELECT *  
2 FROM revenue_csv;
```

Query results

Query status

✔ Completed

Time in queue: 105 ms

Run time

Results (10)

Q Search rows

#	▼	year	▼	revenue	▼	profits
1		2027		643889		45079
2		2023		834658		98764
3		2024		25343		8906
4		2025		123678		82632
5		2026		258000		32424
6		2027		678900		56778
7		2028		980000		576876
8		2029		90000		9000
9		2030		100000		10000
10		2031		345000		75655